Model Insurance: Preliminary Results

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In short. A small edit to a trained model can change what it does. A stored hash detects the edit, but undoing it usually means keeping a full clean copy. Model Insurance asks how little protected information is enough to undo any edit from a given class. I call that amount the *price*. For two simple settings, sparse edits to the biases of a one-hidden-layer ReLU network and sparse edits to the weights of a final linear layer, three things hold. The price is set by how many colors a certain graph needs. It grows with the size of the edit and only logarithmically with the size of the model. And because models compute, a record of the original model's outputs lowers the price, sometimes all the way to zero.

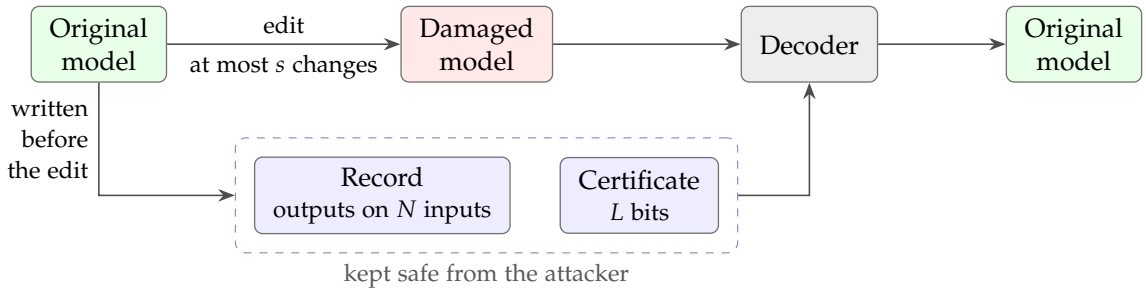


Figure 1. The setting. Before any edit, a record of the original model's outputs and an L-bit certificate are stored where the attacker cannot change them. After an edit from the allowed class, the decoder sees only the damaged model, the record and the certificate, and must return the original. The price is the smallest L that always works.

1 The question

Small edits can change what a model does. Erasing a single direction from a language model's weights disables its refusals [1], and flipping 13 of a ResNet-18's 93 million stored bits drops its top-1 accuracy from 69.8% to 0.1% [2]. A hash of the weights tells us that something changed, but not how to change it back. A clean copy can undo any edit, but it costs as much as the model, and it assumes a copy exists. If the training data and a deterministic training run were kept, retraining would also restore the model for free, so the question matters when some state cannot be recomputed, such as training randomness or on-device personalization.

Model Insurance stores a short certificate before any damage happens (Figure 1). The setting has four parts.

- The *original model*, which we want back.
- An *edit class*, for example "at most s stored numbers change". The attacker may pick any edit in the class and may know the original, the record, the certificate and the scheme.
- A *record*, the original model's outputs on N inputs. Unlike a file, a model computes, so its outputs carry information about its weights.

- A *certificate* of L bits, computed from the original model before the edit.

The record and the certificate are kept where the attacker cannot change them. After the edit, a decoder sees only the damaged model, the record and the certificate, and must return the original. That can mean the exact original, with every stored number restored, or a model whose behavior is within a chosen tolerance of the original. The *price* is the smallest L for which some certificate works against every edit in the class. When the record's inputs are random, "works" means with probability at least $1 - \delta$.

2 The price is a coloring number

Call two possible originals *confusable* if they agree on the record, differ by more than the tolerance, and one edit can turn each of them into the same damaged model. Draw a graph with a node for every original that fits the record and an edge between every confusable pair. If two connected originals received the same certificate, the decoder would face identical inputs in both cases and would fail on one of them. If connected originals always receive different certificates, the decoder can always tell which original it faces. So a certificate is a coloring of this graph, and for exact repair

$$\text{price} \;=\; \bigl\lceil \log_2(\text{number of colors the graph needs}) \bigr\rceil .$$

For repair up to a tolerance r, the same formula brackets the price. The graph that joins confusable originals more than $2r$ apart gives a lower bound, and the one that joins those more than r apart gives an upper bound.

Example. Suppose the record leaves 6 binary weights undetermined and an edit may flip at most one of them. Two originals are confusable exactly when they differ in one or two places, since one flip to each can meet in the middle. A word and its 6 neighbors are pairwise confusable, so at least 7 colors are needed, which is 3 bits. The 3-bit syndrome of a Hamming code gives every confusable pair different labels, so 3 bits also suffice. A full copy would cost 6 bits.

The coloring view is classical. It is Witsenhausen's zero-error problem with side information at the decoder [3], and storing a syndrome so that a corrupted copy can be corrected is the "secure sketch" of Dodis et al. [4]. What is specific to models is the side information. The decoder holds the original model's outputs as well as the damaged weights, and those outputs rule out originals and remove edges from the graph. The rest of this note is about how much they remove.

3 The price grows with the edit, not the model

Let s be the number of stored numbers an edit may change, and m the number of weights the record leaves undetermined. For binary weights, the number of colors lies between the sizes of the Hamming balls of radius s and $2s$ in $\{0,1\}^m$, so the price is about $s \log_2(m/s)$ bits, up to a constant factor. With real-valued weights repaired up to a tolerance, a precision term is added, roughly s times the logarithm of the weight range divided by the tolerance, and matching upper and lower bounds hold up to constants. Even locating an edit without undoing it can cost $\log_2 \binom{m}{s} \geq s \log_2(m/s)$ bits.

For scale, take the output layer of an 8-billion-parameter language model, a $128{,}256 \times 4{,}096$ matrix of 525 million 16-bit weights, and allow any 64 of them to change, with no record at all. A

Reed–Solomon syndrome of 128 numbers modulo the prime 525,336,587 undoes every such edit with 3,712 bits, or 464 bytes. Any certificate needs at least 2,582 bits in this setting. A clean copy takes about a gigabyte.

4 Records lower the price, sometimes to zero

Shallow ReLU biases. Consider a one-hidden-layer ReLU network with q hidden neurons in which only the hidden biases can be edited. Suppose the inputs include, for each neuron, an isolating region where only that neuron is active. One recorded output from neuron j's region reveals neuron j's bias exactly, so the certificate only has to cover the neurons whose regions the record missed. For example, let 3 of 10 biases be secretly switched on, and let the edit switch all of them off. If the record visits 4 regions and finds 1 of the 3, then 15 originals remain possible, so 4 bits suffice where a full copy of the pattern would take 7. If each isolating region has probability α/q, then after about $(q/\alpha)\ln(q/\delta)$ random inputs the record has visited every region with probability at least $1-\delta$, and the price is zero.

Final linear layer, exact outputs. Now let the final layer be editable and the features feeding it be intact. If its outputs are recorded exactly on $N \geq 2s$ inputs whose features are in general position, no two layers within s changes of the same damaged layer share a record. The record alone pins the layer down, and the price is zero.

Final linear layer, rounded outputs. Real records are rounded. Consider one row of the final layer, the d weights that produce one output. Suppose each recorded output is only the index of a cell of width W, the weights sit on a grid of step Δ, and one weight in the row may change. Together, the N records narrow the weights to a window of about $W/(N\Delta)$ grid points, and one change can confuse originals that differ in two weights, so the certificate has to name a point in a two-dimensional window. With Gaussian features and a randomly shifted rounding grid, two checksums modulo a prime of order $\frac{W}{N\Delta}\log\frac{d}{\delta}$ suffice, and a matching lower bound holds. So, while the record is coarse compared with the weight grid, the price is about

$$2\log_2\frac{W}{N\Delta}\ \text{bits},$$

up to a term of order $\log\log(d/\delta)$. Each doubling of the record saves about two bits, until the record alone suffices. With k changes in the row, the leading term becomes $2k\log_2\big(W/(N\Delta)\big)$. This guarantee is for the model actually being protected, with the random shift drawn after the model is fixed. If the certificate must instead work for every model that could have produced the same record, the record helps far less. When a row has more weights than there are records, the price is then about $2\log_2(W/\Delta) - \log_2\ln N$.

Same statistics, different price. Which inputs enter the record matters, not only how many or how varied they are. Table 1 compares two 2×2 feature matrices with the same Gram matrix. Under H_1, only the original fits the record, so repair is free. Under H_2, five originals fit it, and any two of them can be turned into the same damaged model by one change each, so the price is 3 bits.

Table 1. Same Gram matrix, different price. The original weights are $(0, 0)$, weights lie in $\{-2, \ldots, 2\}$, outputs are recorded as $\lfloor (Ha + 25)/50 \rfloor$, and one weight may change.

Features H	$H^\top H$	Originals that fit the record	Price
$H_1 = \begin{pmatrix} 30 & 0 \\ 0 & 30 \end{pmatrix}$	$900\, I$	$(0, 0)$	0 bits
$H_2 = \begin{pmatrix} 18 & 24 \\ 24 & -18 \end{pmatrix}$	$900\, I$	$(0, 0),\ (\pm 1, 0),\ (0, \pm 1)$	3 bits

5 What comes next

These are the simplest settings. Four directions come next.

- *Deep networks*, where an edit's effect compounds across layers. For general shallow ReLU networks without isolating regions, the technical notes give conditions under which the record alone suffices, but not yet the price.
- *Dense edits that carry little information.* The ablation that disables refusals changes every weight in the matrices it edits, yet each change is rank one. Its price should be set by that information, not by the number of weights it touches.
- *Efficient decoding.* The bounds here count bits. Some constructions, such as Reed–Solomon syndromes, decode efficiently, but finding the original is not known to be efficient in general.
- *Training for insurability.* An edit is expensive to insure when it can hide from the record, changing behavior only on inputs the record never saw. Training that leaves equally accurate models with fewer such hiding places would make repairability a design goal, like robustness.

Scope. The results are information-theoretic and assume the record and the certificate are safe from the attacker. Some rely on strong assumptions, namely isolating input regions for the ReLU results and Gaussian features with a randomly shifted rounding grid for the rounded-record results. Full statements and proofs are in two technical notes, available on request.

References.

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